Big Data (6CS030)

Workshop 2:
Report on Subject Specific Big Data Process

Topic: Water Quality Analysis and Potability Prediction

Student Name: Rakshyak Basyal

Student ID: 2432209

Tutor: Ms. Rakhee Pandey

Lecturer: Mr. Sandesh Lamsal

Table of Contents

1. Introduction 1

2. Step 1: Data Source Identification and Acquisition 1

2.1 Identified Data Source 1

2.2 Insight Generation Goals 2

2.3 Data Acquisition 2

3. Step 2: Data Exploration, Understanding, Transformation and Model Training 3

3.1 Data Understanding 3

3.2 Missing Value Handling 3

3.3 Data Transformation 3

3.4 Exploratory Data Analysis (EDA) 4

3.5 Insight-Oriented Grouped Analysis 9

3.6 Model Training 9

4. Step 3: Insights Generated 9

4.1 Model Performance 9

4.2 Feature Importance 10

5. Step 4: Deployment Plan 12

6. Conclusion 13

1. Introduction

Access to safe and clean drinking water is one of the most fundamental requirements for human health and well-being. According to the World Health Organization (WHO), contaminated water is responsible for a significant number of preventable diseases globally each year. Assessing water quality, however, involves measuring and interpreting a wide range of physicochemical parameters, which makes manual evaluation time-consuming and prone to error.

This report presents a big data process applied to the domain of water quality analysis. The project uses the Water Potability Dataset from Kaggle, which contains 3,276 water sample records with nine physicochemical measurements, to explore patterns in water quality data and build a machine learning model capable of predicting whether a given water sample is safe for human consumption.

The report is structured around four key steps as defined by the Workshop 2 guidelines. Step 1 covers the identification of the data source and the approach for data acquisition. Step 2 details the data exploration, cleaning, transformation, and model training process with full working code in Python via Google Colab. Step 3 summarizes the insights generated from both the exploratory analysis and the predictive model. Step 4 outlines a realistic deployment plan for making the model and its predictions accessible to end users.

2. Step 1: Data Source Identification and Acquisition

2.1 Identified Data Source

The dataset used for this project is the Water Potability Dataset, publicly available on Kaggle. The dataset contains 3,276 water sample records across 10 attributes that measure various physicochemical properties of water, along with a binary target variable indicating whether the water is safe to drink or not.

The features in the dataset are:

- pH – Measures the acidity or alkalinity of water (safe range: 6.5 to 8.5)
- Hardness – Amount of dissolved calcium and magnesium compounds
- Solids – Total dissolved solids (TDS) in water
- Chloramines – Amount of chloramines used for disinfection
- Sulfate – Dissolved sulfate concentration
- Conductivity – Electrical conductivity of water
- Organic Carbon – Amount of organic carbon from natural and synthetic sources

- Trihalomethanes – Chemicals formed during water disinfection
- Turbidity – Measure of cloudiness of water
- Potability – Target variable (1 = Potable, 0 = Not Potable)

2.2 Insight Generation Goals

The dataset can be used to generate several valuable insights related to water quality. By studying the relationship between chemical and physical properties of water and the potability label, it becomes possible to identify which factors are most associated with safe or unsafe water. The analysis can help answer questions such as whether pH levels differ between potable and non-potable water, whether sulfate or solids are linked to poor water quality, and which features have the strongest influence on classification.

The data can therefore be used for:

- understanding water quality patterns,
- comparing potable and non-potable water samples,
- identifying key indicators of water safety,
- supporting predictive analysis of potability.

These insights may be useful in environmental monitoring, laboratory testing, municipal water management, and early warning systems for unsafe water conditions.

2.3 Data Acquisition

The dataset was manually downloaded from Kaggle (Water Potability Dataset by Aditya Kadiwal) (Access Link: https://www.kaggle.com/datasets/adityakadiwal/water-potability) as a CSV file and loaded into Google Colab via Google Drive.

Although the current dataset is provided as a CSV file for academic analysis, in a real-world big data process the acquisition of water quality data could be automated. Water data may be collected from:

- IoT-based water quality sensors,
- environmental monitoring stations,
- laboratory information systems,
- water treatment plants,
- or public environmental databases.

A practical automated data acquisition process could follow an ETL pipeline. First, raw readings would be extracted from sensors, APIs, or databases. Then the data would be transformed into a standard structured format. Finally, the processed data would be loaded into a central storage system such as a relational database, cloud storage environment, or data lake.

Automation could be implemented using Python scripts scheduled through Cron, Windows Task Scheduler, or orchestration tools such as Apache Airflow. Such a system would enable periodic collection of freshwater quality measurements, allowing continuous monitoring and analysis without manual file handling.

3. Step 2: Data Exploration, Understanding, Transformation and Model Training

All data exploration, transformation, and model training tasks were carried out in Python using Google Colab. Below is a summary of each stage along with the charts produced and their analytical interpretations.

3.1 Data Understanding

The dataset contains 3,276 rows and 10 columns. Inspection revealed missing values in three columns — ph (491 missing), Sulfate (781 missing), and Trihalomethanes (162 missing). No duplicate rows were found. Summary statistics showed that features span very different ranges — for example Solids goes up to 60,000+ while Turbidity stays within a small range.

3.2 Missing Value Handling

Missing values in ph, Sulfate, and Trihalomethanes were filled using the median of each column. The median was chosen over the mean because it is more robust to outliers, which is important for skewed water quality data.

3.3 Data Transformation

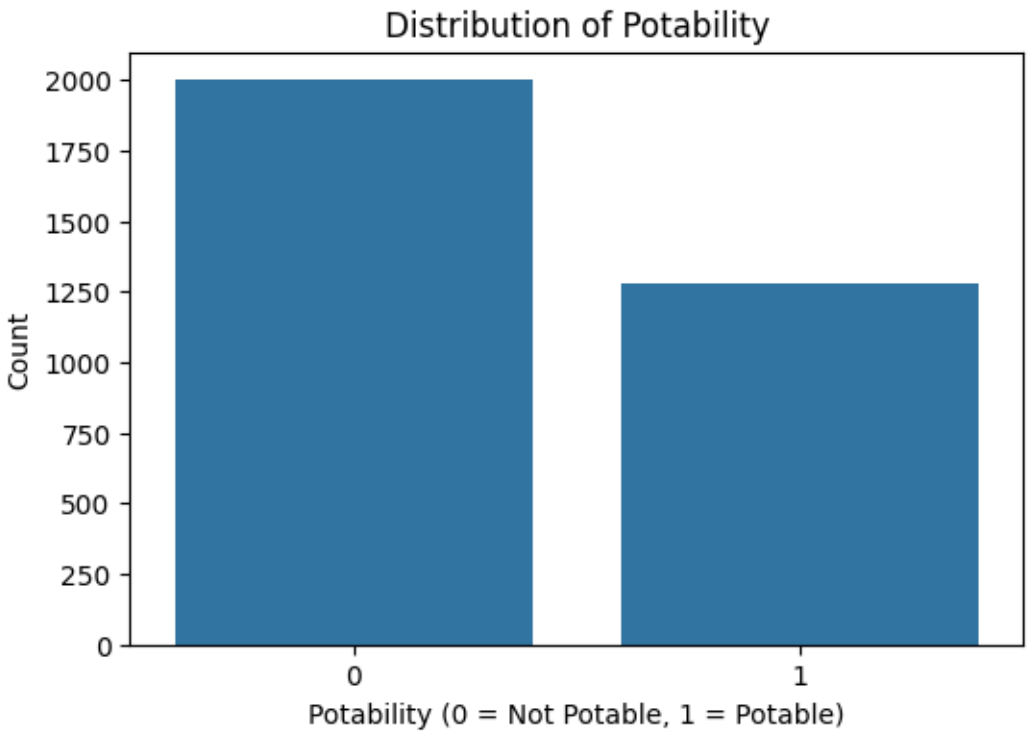
Three new categorical features were derived:

- ph_category – Acidic (< 6.5), Neutral (6.5–8.5), Alkaline (> 8.5) based on WHO guidelines
- hardness_category – Low, Medium, High using quantile-based binning
- solids_category – Low, Medium, High using quantile-based binning

3.4 Exploratory Data Analysis (EDA)

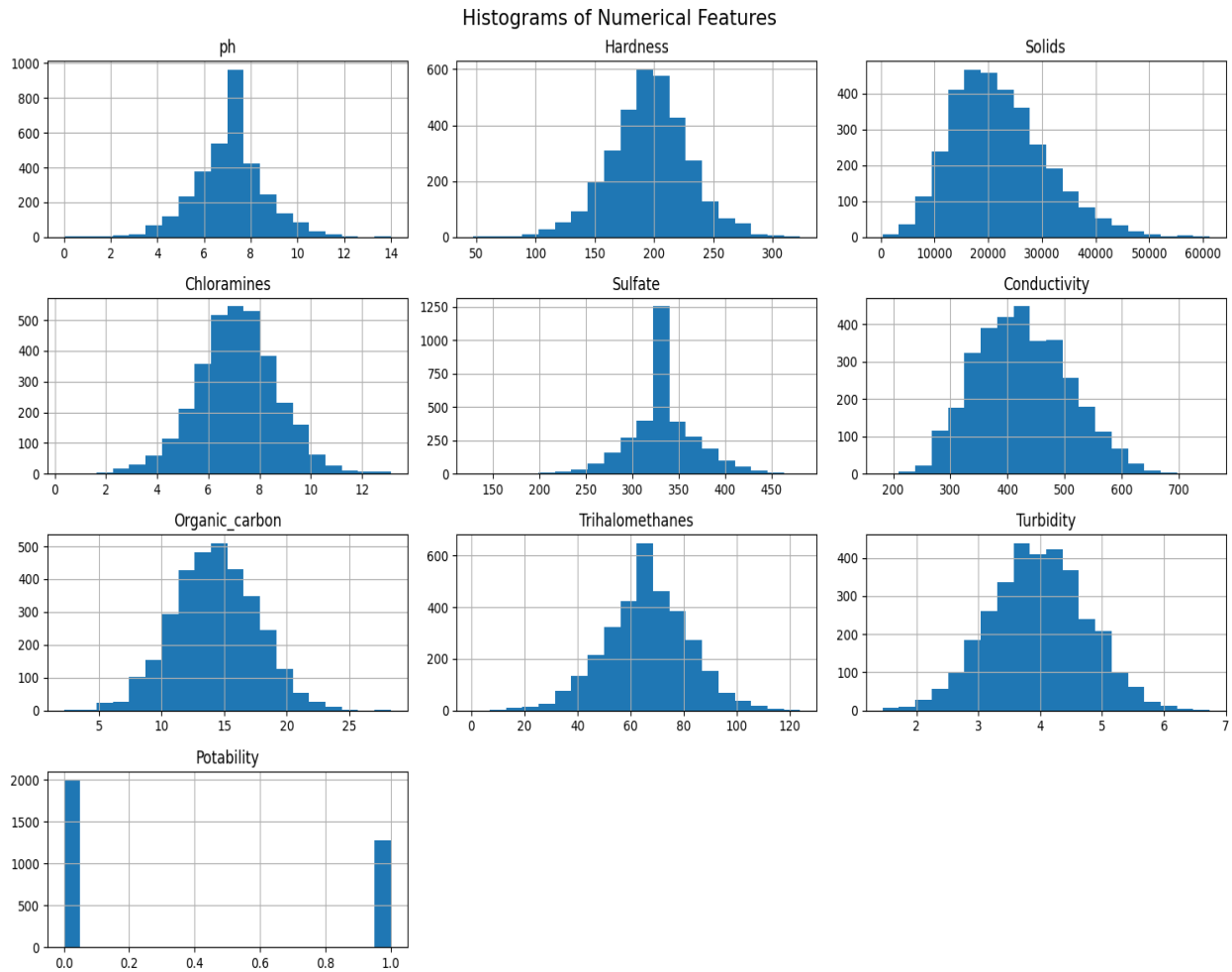
Distribution of Potability

The count plot of the Potability variable showed a class imbalance in the dataset. Approximately 61% of samples are classified as Not Potable (0) and 39% as Potable (1). This imbalance is important to note as it can affect model performance and evaluation metrics, and explains why accuracy alone is not a sufficient evaluation metric.



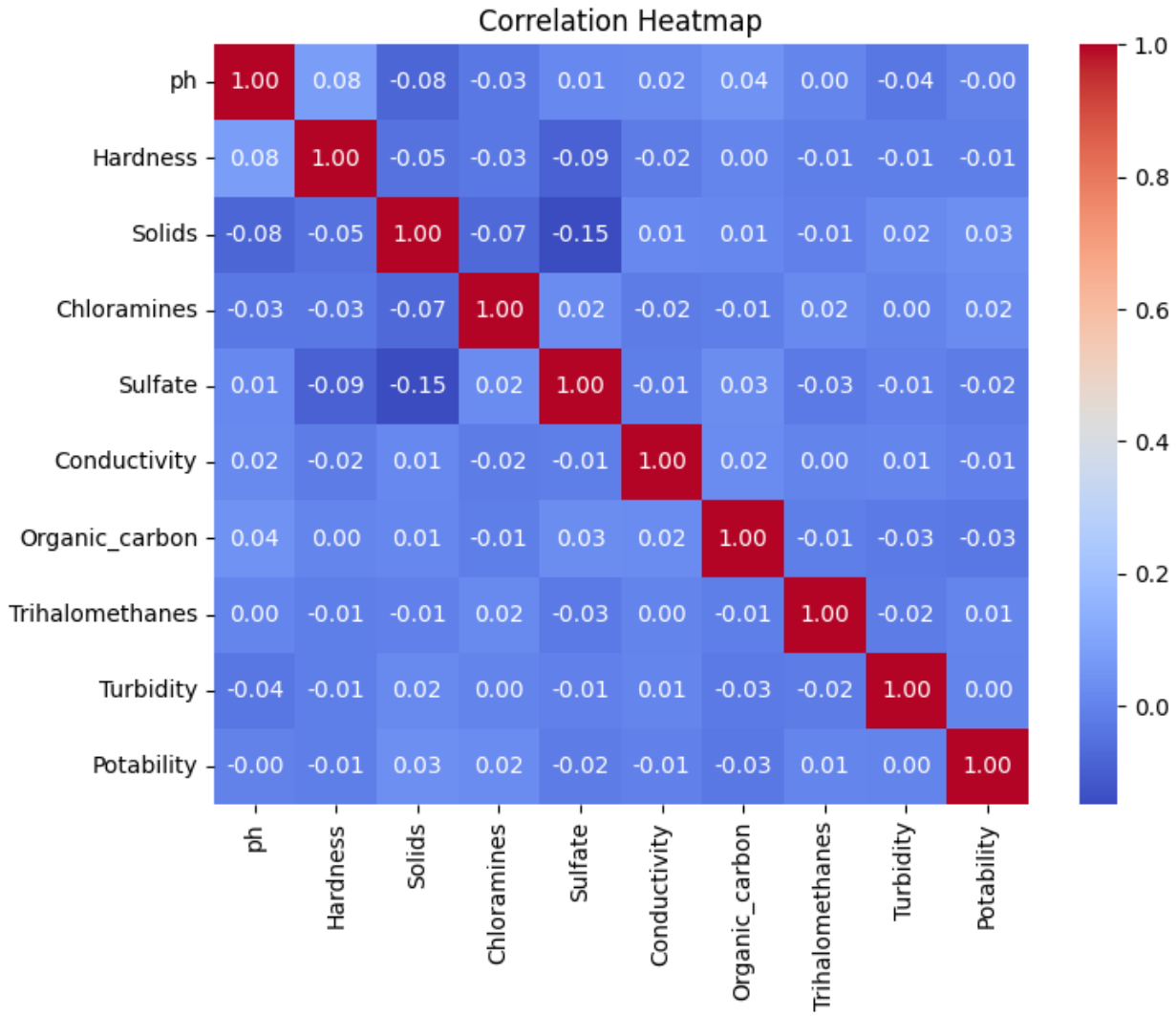
Histograms of Numerical Features

The histograms of all numerical features revealed that most features follow an approximately normal distribution. Solids showed a slight right skew, suggesting the presence of a few high-TDS water samples. pH, Hardness, Chloramines, Conductivity, Organic Carbon, Turbidity, and Trihalomethanes all appeared approximately normally distributed, with no extreme outliers that would require removal.



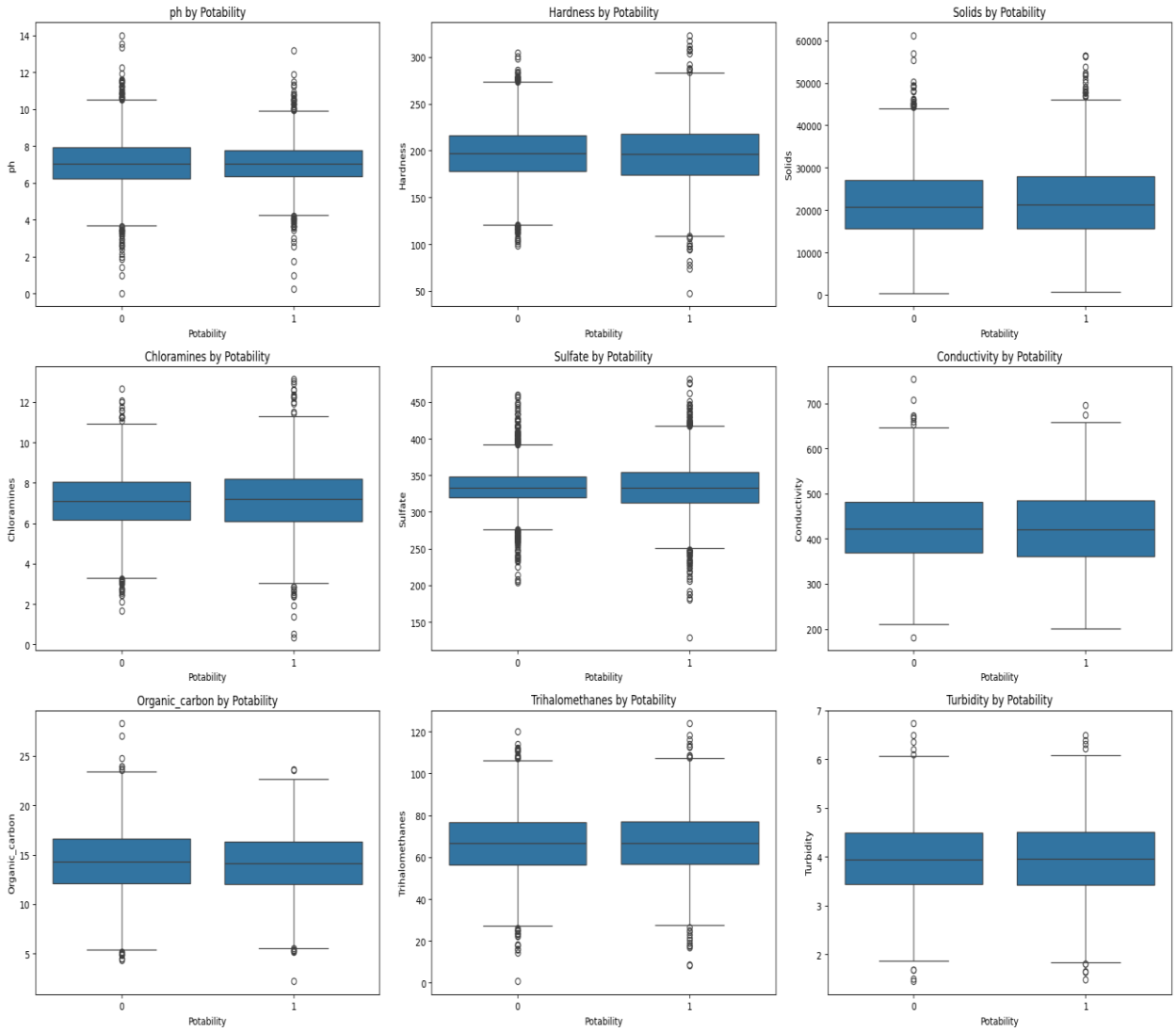
Correlation Heatmap

The correlation heatmap was generated to examine relationships between all numerical features. The results revealed very low correlations between most feature pairs, with no pair exceeding an absolute correlation of 0.1. This indicates that the features are largely independent of each other, which is beneficial for model training as it reduces the risk of multicollinearity. Importantly, no single feature showed a strong linear correlation with the Potability target variable, suggesting that potability is driven by complex, non-linear interactions among multiple features rather than any single dominant predictor.



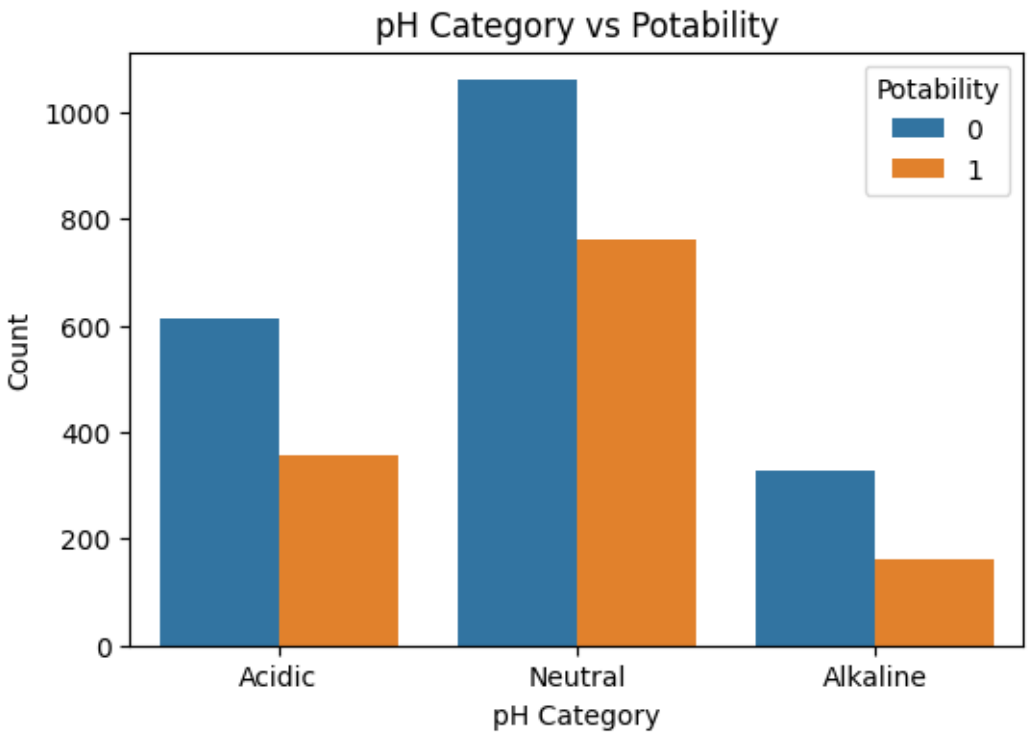
Boxplots by Potability

The boxplots comparing feature distributions across Potable (1) and Not Potable (0) samples showed that the median values of most features are very similar between the two classes. There is substantial overlap in the interquartile ranges across all features. This confirms the finding from the correlation heatmap that no single feature alone is a strong discriminator for potability. However, subtle distributional differences across features collectively contribute to the model's predictive ability.



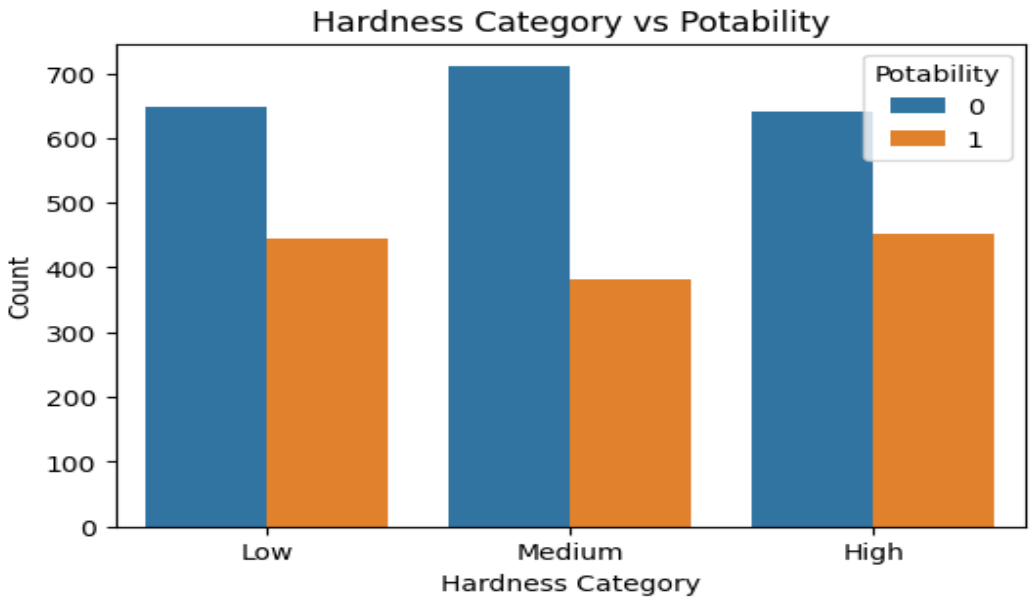
pH Category vs Potability

The plot of pH categories against potability showed that water samples across all three pH categories (Acidic, Neutral, Alkaline) contain both potable and non-potable samples. Neutral pH water had the highest count of potable samples, consistent with WHO guidelines that recommend a pH range of 6.5 to 8.5 for safe drinking water. However, the difference is not dramatic, reinforcing that pH alone is not sufficient to determine potability.



Hardness and Solids Category vs Potability

The hardness and solids category plots similarly showed no strong relationship between these categories and potability. All hardness and solids levels had comparable proportions of potable and non-potable samples. This further supports the conclusion that potability is determined by the combined effect of all water quality parameters rather than any individual measure.



3.5 Insight-Oriented Grouped Analysis

Mean feature values were computed for each potability class. The comparison revealed marginal differences in mean values across features between potable and non-potable water. Potability rates were also computed by pH, hardness, and solids categories. Neutral pH water had a slightly higher potability rate compared to acidic or alkaline water, which aligns with established water quality standards.

3.6 Model Training

A Random Forest Classifier was trained to predict water potability. Random Forest was selected due to its ability to capture complex, non-linear relationships between features and its robustness to feature scale differences. The dataset was split into 80% training and 20% testing sets using a fixed random state of 42 for reproducibility.

The model was trained with 100 decision trees (n_estimators=100). After training, predictions were made on the test set and the model was evaluated using accuracy score, classification report, and a confusion matrix heatmap.

4. Step 3: Insights Generated

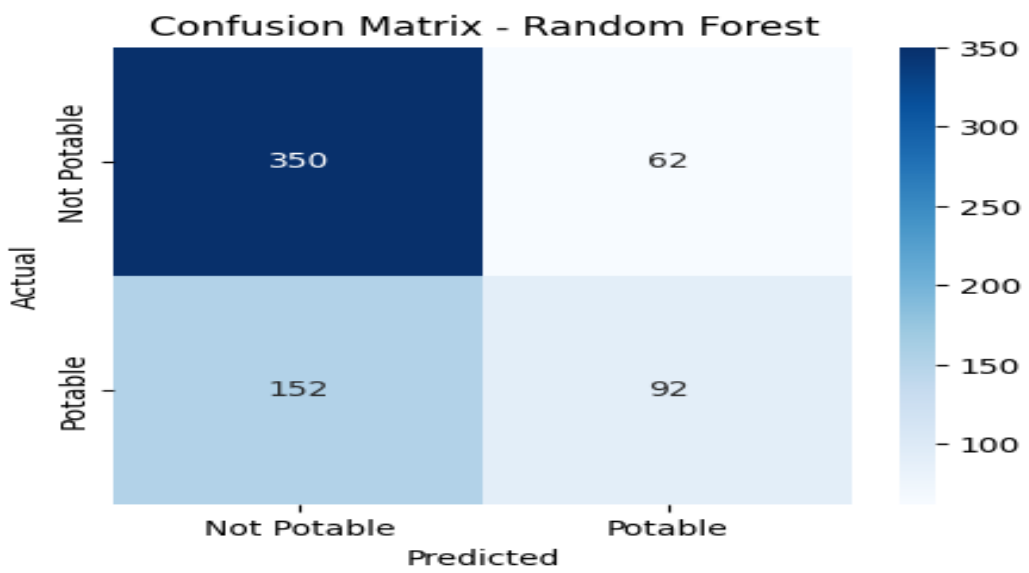
4.1 Model Performance

The Random Forest model achieved an overall accuracy of 67.4% on the test set. The detailed classification report is presented below:

Class	Precision	Recall	F1-Score	Support
Not Potable (0)	0.70	0.85	0.77	412
Potable (1)	0.60	0.38	0.46	244
Accuracy			0.67	656
Macro Avg	0.65	0.61	0.61	656
Weighted Avg	0.66	0.67	0.65	656

The model performs better at identifying non-potable water (F1-score: 0.77) compared to potable water (F1-score: 0.46). The lower recall for the Potable class (0.38) indicates that the model misses a significant proportion of actually safe water samples, predicting them as unsafe. This is partly attributed to the class imbalance in the dataset (61% non-potable vs 39% potable) and the absence of strong individual feature predictors as established during EDA.

While 67.4% accuracy is modest, it demonstrates that the combination of all nine water quality parameters carries meaningful predictive signal. Further improvement could be achieved through techniques such as SMOTE for class balancing, hyperparameter tuning, or using more advanced ensemble methods.



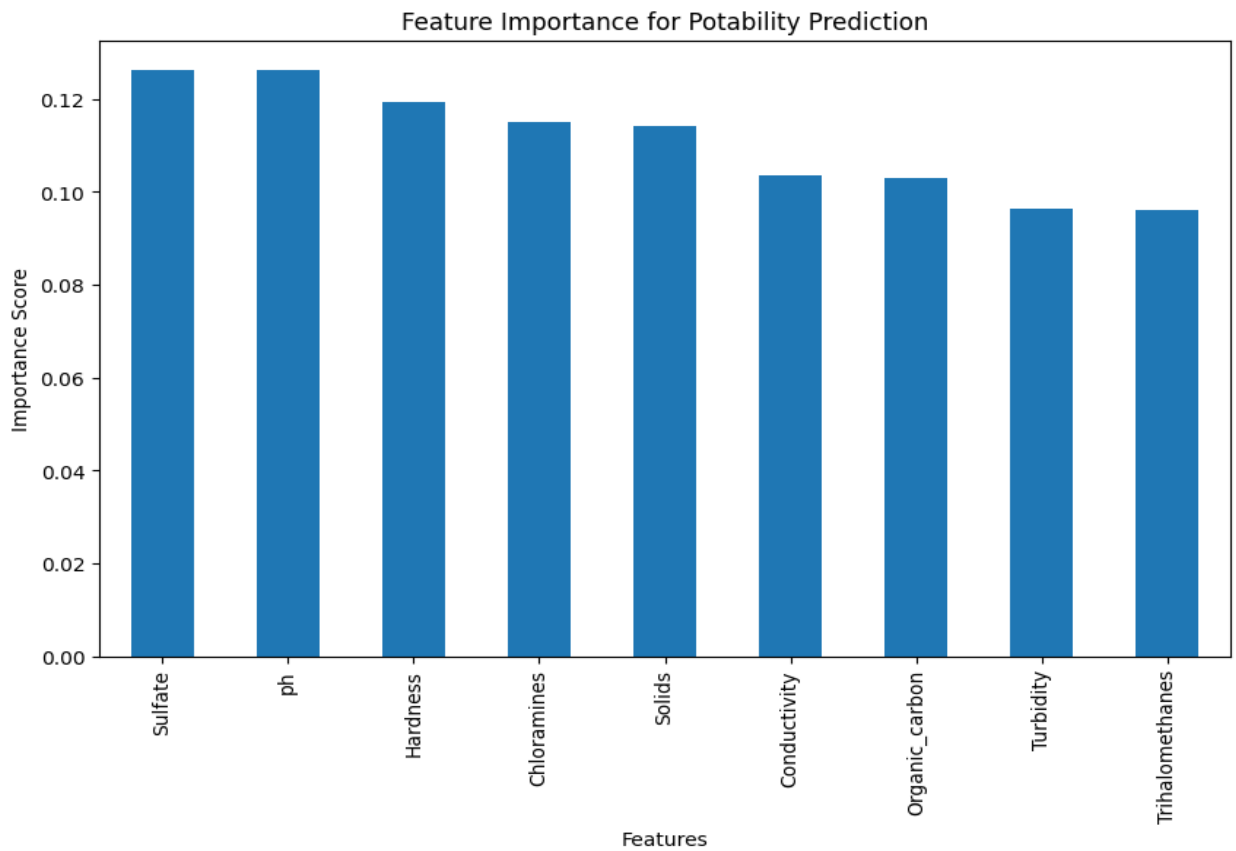
4.2 Feature Importance

The feature importance scores derived from the Random Forest model are presented below:

Rank	Feature	Importance Score
1	Sulfate	0.1263
2	pH	0.1261
3	Hardness	0.1193
4	Chloramines	0.1151

Rank	Feature	Importance Score
5	Solids	0.1142
6	Conductivity	0.1034
7	Organic Carbon	0.1029
8	Turbidity	0.0965
9	Trihalomethanes	0.0960

The feature importance scores are remarkably evenly distributed across all nine features, with Sulfate (12.63%) and pH (12.61%) being the top two contributors. This near-uniform distribution of importance confirms the EDA finding that no single feature dominates potability prediction. Instead, all physicochemical parameters collectively contribute to determining water safety. This has an important practical implication: water quality monitoring systems should measure all parameters comprehensively rather than relying on just one or two indicators.



5. Step 4: Deployment Plan

The developed process can be deployed as a practical water quality monitoring and analytics system. In a real deployment scenario, water quality measurements would be collected automatically from sensors, testing devices, or laboratory systems and transferred into a centralized platform.

The deployment architecture could include the following stages. First, data acquisition would take place through automated ingestion from monitoring stations or APIs. Second, the incoming data would pass through preprocessing steps such as missing value handling, validation, and formatting. Third, the cleaned data would be stored in a cloud database or data warehouse. Fourth, an analytics layer would generate charts, alerts, and summary reports for users.

A dashboard-based deployment would be especially useful. Such a dashboard could display current water quality indicators, recent trends, risk alerts, and potability predictions. Authorities or organizations could use this dashboard to monitor water conditions continuously and respond more quickly to abnormal values.

If the optional predictive model is included in the deployment, it could be integrated into the pipeline to flag samples that appear likely to be non-potable. However, because the model performance is moderate, it should be used only as a supplementary tool alongside standard water testing procedures.

This process could be deployed using modern data tools such as Python-based ETL workflows, cloud storage, API integration, and visualization platforms including Power BI, Tableau, or a custom web dashboard. In this way, the analytical workflow developed in this workshop could be scaled into a real-world environmental data monitoring solution.

6. Conclusion

This report presented a detailed subject-specific big data process for water quality analysis and potability prediction. The process began by identifying a suitable dataset and explaining how automated acquisition could be implemented in a real-world scenario. It then demonstrated data exploration, missing value handling, transformation, and optional model training using Python.

The analysis showed that the dataset contained important preprocessing challenges, especially missing values in several chemical indicators. After cleaning and transformation, exploratory data analysis was used to understand variable distributions, compare potable and non-potable samples, and identify meaningful patterns. The optional Random Forest model achieved moderate predictive performance and highlighted sulfate, pH, hardness, chloramines, and solids as influential features.

Overall, the project demonstrated that a structured big data process can be used to transform raw environmental data into useful analytical insights. Although the predictive model should not be treated as a fully reliable standalone system, the complete workflow provides a valuable foundation for water quality monitoring, insight generation, and future deployment in practical settings.